Nearest neighbors, matrix product, matrix factorization

Lecture notes. AI in Industry, ETH Zurich, weekend 2, Friday 13 November 2026, 08:15 to 12:00.

1 The inner product

Rating. A number from -2 to 2: 2 loved, 1 liked, 0 neutral, -1 disliked, -2 hated. A rating of 1 to 5 stars becomes the number of stars minus 3.

Customer. A row of ratings, one per film.

Inner product. Of two rows of the same length, $a = (a_1, \ldots, a_n)$ and $b = (b_1, \ldots, b_n)$: the sum of the products of their entries,

$$a \cdot b = a_1 b_1 + a_2 b_2 + \cdots + a_n b_n.$$

One film. The product of two customers' ratings says how much they agree: positive when both are on the same side, negative when they are on opposite sides, 0 when one of them is neutral, further from 0 when the opinions are stronger. Jaws: Ivo and Pia, $2 \times 2 = 4$; Ivo and Rosa, $2 \times (-2) = -4$; Pia and Kai, $2 \times 1 = 2$; Kai and Tom, $1 \times (-1) = -1$.

Several films. The inner product adds the agreements of all films. Vera and Kai:

	Jaws		The Shining		Notting Hill		
Vera	2		1		-1		
Kai	1		2		-2		
	2	+	2	+	2	=	6

With ratings 1 and -1 only, the inner product is the number of matches minus the number of mismatches.

Game: Once Upon a Video Store, acts 1 to 3.

2 The nearest neighbor classifier

Classifier. A rule that predicts the class of a new case. Here the case is a newcomer, and the class is her rating of a film she has not seen.

Nearest neighbor classifier.

1. Similarity: the inner product of the newcomer's row with each regular customer's row, on the films the newcomer rated.
2. Nearest neighbor: the regular with the largest inner product.
3. Prediction: the nearest neighbor's rating of the film.

k **nearest neighbors.** The k regulars with the largest inner products vote with their ratings of the film. The prediction is the rating with the most votes; on a tie, the nearer neighbor's vote wins.

Example. Vera rated three films. The regulars, sorted by their inner product with her:

	Jaws	The Shining	Notting Hill	Love Actually	inner product with Vera
Vera	2	1	−1	?	
Kai	1	2	−2	−2	6
Pia	2	1	2	2	3
Ivo	2	−2	0	1	2
Jana	0	1	0	0	1
Tom	−1	−2	−1	−1	−3
Rosa	−2	−1	2	2	−7

Nearest neighbor: Kai, 6. Prediction for Love Actually: −2, Kai's rating. Vera's rating after watching it: −2.

A count of exact matches picks Pia (two films) over Kai (none); Pia's rating of Love Actually is 2. The inner product weighs each agreement by the strength of both ratings.

Game: Once Upon a Video Store, act 4. Notebook: nearest neighbors.

3 The matrix product

Matrix. A table of numbers. Its size is $n \times m$: n rows, m columns.

Profiles. Each customer answers three questions with a number from −2 to 2: real story (2) or fiction (−2), comedy (2) or drama (−2), action (2) or calm (−2). A film critic answers the same questions for every film. P holds the customers' answers, one row per customer. Q holds the critic's answers, one column per film.

Embedding. The numbers that describe a customer (a row of P) or a film (a column of Q).

Score. The inner product of a customer's row and a film's column. The higher the score, the better the film suits the customer.

Matrix product. P of size $n \times k$ times Q of size $k \times m$, defined when P has as many columns as Q has rows. PQ has size $n \times m$. Its entry in row i and column j is the inner product of row i of P and column j of Q. With Q drawn above PQ and P on its left, each entry sits where its row and its column meet:

	The Intouchables	Apollo 13	Men in Black	Interstellar
real	2	2	−2	−2
comedy	1	−2	1	−2
action	−1	1	2	0

Q

	real	comedy	action
Sara	2	1	−2
Leo	−2	2	2
Maya	1	−2	1

P

7	0	−7	−6
−4	−6	10	0
−1	7	−2	2

PQ

Sara and The Intouchables: $2 \times 2 + 1 \times 1 + (-2) \times (-1) = 4 + 1 + 2 = 7$. Each customer's highest score: Sara, The Intouchables (7); Leo, Men in Black (10); Maya, Apollo 13 (7).

Sizes: 5×4 times 4×6 gives 5×6. The product of 2×3 and 4×2 is undefined: 3 columns against 4 rows.

Game: Matchmaker, part one. Notebook: matrix factorization, part 1.

4 Matrix factorization

4.1 The problem

Matrix factorization. Given only the ratings R, n customers by m films, find P ($n \times k$) and Q ($k \times m$) whose product PQ is close to R. The number k is chosen in advance. The rows of P and the columns of Q are embeddings; their numbers have no names.

Prediction. The entry of PQ for a customer and a film.

Loss. Of one rating: $(\text{prediction} - \text{rating})^2$.

Average loss. The average of the losses of the known ratings. Matrix factorization looks for the P and Q with the smallest average loss.

Example: three customers, two films, $k = 2$, a first guess of P and Q:

Q

	Paddington	The Martian
1	−1	−2
2	2	−1

P

	1	2
Ida	−1	−2
Raj	−2	0
Elsa	2	2

predictions PQ

	−3	4
	2	4
	2	−6

ratings R

Paddington	The Martian
2	1
−1	2
1	−2

losses

25	9
9	4
1	16

Average loss: $(25 + 9 + 9 + 4 + 1 + 16)/6 = 64/6 = 10.67$.

4.2 One number at a time

For one number w of P or Q:

1. Before: the average loss.
2. After: the average loss with w changed to $w + 0.01$.
3. Slope: $\nabla = (\text{after} - \text{before})/0.01$.
4. Update: $w \leftarrow w - \alpha\nabla$, with the step size $\alpha = 0.1$.

A negative slope moves w up, a positive slope moves it down. A round updates every number once, the numbers of P first. Rounds repeat until the average loss is below 0.0001.

Example: The Martian's second number, $w = -1$. Before: 10.667. After, with $w = -0.99$: 10.620. Slope: $\nabla = -0.047/0.01 = -4.7$. Update: $w \leftarrow -1 - 0.1 \times (-4.7) = -0.53$; the average loss falls to 8.78. After 99 rounds the average loss is below 0.0001 and every prediction is within 0.02 of its rating.

Game: Matchmaker, part two. Notebook: matrix factorization, part 2.

4.3 Gradient descent

Slope rule. For a number w and a known rating whose prediction contains w: $2 \times$ partner $\times$ (prediction $-$ rating), where the partner is the number that w multiplies in that prediction. A rating whose prediction does not contain w gives 0.

Slope of the average loss. The average of the slope rule over the known ratings. The slope measured with a change of 0.01 is close to it.

Gradient. The slopes of all numbers of P and Q.

Gradient descent. Every number moves at once, $w \leftarrow w - \alpha \times$ slope, again and again until the average loss stops falling.

Example: from the first guess above, with $\alpha = 0.2$, one step takes the average loss from 10.67 to 1.97, and 52 steps take it below 0.0001.

Game: Matchmaker, optional part. Notebook: matrix factorization, optional, parts 3 and 4.

4.4 Predicting unknown ratings

The average loss uses the known ratings only. After the descent, PQ has an entry for every customer and every film; for an unknown rating, that entry is the prediction.

Example: real ratings from MovieLens[1], 10 users and 12 films, each rating the number of stars minus 3, three ratings hidden. Gradient descent with $k = 2$, $\alpha = 3$ and 200 steps takes the average loss of the 117 known ratings from 1.84 to 0.25. The hidden ratings and their predictions:

Film	MovieLens user	Prediction	Rating
Independence Day	64	0.67	0
Braveheart	28	0.62	0.5
Seven	239	1.98	2

Game: Matchmaker, part three.

5 Games and notebooks

Games, on the course site:

Once Upon a Video Store	https://eth-ainit-hs26.github.io/AINIT-PAGE/viz/we2/video-store/
Matchmaker, part one	https://eth-ainit-hs26.github.io/AINIT-PAGE/viz/we2/matchmaker-part-1/
Matchmaker, part two	https://eth-ainit-hs26.github.io/AINIT-PAGE/viz/we2/matchmaker-part-2/
Matchmaker, part three	https://eth-ainit-hs26.github.io/AINIT-PAGE/viz/we2/matchmaker-part-3/
Matchmaker, optional part	https://eth-ainit-hs26.github.io/AINIT-PAGE/viz/we2/matchmaker-optional/

Notebooks, in the public repository https://github.com/eth-ainit-hs26/w2-cx-public. Each link opens a notebook in Colab:

Nearest neighbors	student, solutions
Matrix factorization, parts 1 and 2	student, solutions
Matrix factorization, optional parts 3 and 4	student, solutions

[1] MovieLens ml-latest-small, GroupLens Research, University of Minnesota. F. Maxwell Harper and Joseph A. Konstan, The MovieLens Datasets: History and Context, ACM Transactions on Interactive Intelligent Systems 5(4), 2015.